

Problem-Solution Fit Analysis

Date	30 June 2026	
Team ID		
Project Name	Rising Waters - Flood Hazard Forecasting System	
Maximum Marks	2 Marks	

1. CUSTOMER SEGMENTS

- Regional meteorologists
- Disaster relief teams
- Evacuation managers
- Local authorities

6. CUSTOMER LIMITATIONS

- Limited bandwidth in storms
- Mobile-only device access
- Time constraints during flood
- No complex terminal tools

5. AVAILABLE SOLUTIONS

- Physics hydrodynamic models
- Manual spreadsheets logs
- Regional warning sirens
- Regional weather forecasts

2. PROBLEMS / PAINS

- Long model compute delays
- Excel files freeze on mobile
- Human typos in input logs
- Unclear advisory recommendations

9. PROBLEM ROOT / CAUSE

- Models require big servers
- Spreadsheets are unvalidated
- No mobile UI validation checks
- No real-time audit tools

7. USER BEHAVIOR

- Consults past warning logs
- Checks cloud density visual
- Enters rainfall data manually
- Broadcasts warnings to centers

3. TRIGGERS TO ACT

- Rapid cloud formations
- Monsoon cycle warning calls
- River gauge limits alarm
- Unexpected cloudburst rain

10. YOUR SOLUTION

- Flask web application
- Decision Tree model split
- AJAX UI form interface
- Vercel serverless deployment

8. CHANNELS OF ACTION

- Evacuation relief radios
- SMS warnings broadcasts
- Mobile portal warnings page
- Local sirens activations

4. EMOTIONS (BEFORE / AFTER)

- Before: Stressed, uncertain about flood time and parameters, dependent on slow hydrodynamic forecasts.
- After: Reassured and confident with zero-latency web warning calculations and clear bulleted Risk Audit checklists.